



Original Article

Robust transformer-based anomaly detection for nuclear power data using maximum correntropy criterion

Shuang Yi^a, Sheng Zheng^{a,b,*}, Senquan Yang^{c,d}, Guangrong Zhou^b, Junjie He^b^a College of Electrical Engineering & New Energy, China Three Gorges University, Yichang, 443002, China^b College of Science, China Three Gorges University, Yichang, 443002, China^c China Nuclear Industry Key Laboratory of Simulation Technology, Wuhan, 443074, China^d China Nuclear Power Operation Technology Corporation, Ltd, Wuhan, 443074, China

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ABSTRACT

Due to increasing operational security demands, digital and intelligent condition monitoring of nuclear power plants is becoming more significant. However, establishing an accurate and effective anomaly detection model is still challenging. This is mainly because of data characteristics of nuclear power data, including the lack of clear class labels combined with frequent interference from outliers and anomalies. In this paper, we introduce a Transformer-based unsupervised model for anomaly detection of nuclear power data, a modified loss function based on the maximum correntropy criterion (MCC) is applied in the model training to improve the robustness. Experimental results on simulation datasets demonstrate that the proposed *Trans*-MCC model achieves equivalent or superior detection performance to the baseline models, and the use of the MCC loss function is proven can obviously alleviate the negative effect of outliers and anomalies in the training procedure, the F1 score is improved by up to 0.31 compared to *Trans*-MSE on a specific dataset. Further studies on genuine nuclear power data have verified the model's capability to detect anomalies at an earlier stage, which is significant to condition monitoring.

1. Introduction

As is well-known, severe nuclear accidents can result in immeasurable losses and long-lasting impacts. Therefore, operational security has always been the top priority for nuclear power plants. The high sensitivity to safety has led to increasing demands for operational status monitoring and the detection of abnormal events [1]. Traditionally, nuclear power plants collect real-time data from various sensors through the supervisory control and data acquisition (SCADA) system [2]. Qualified technical personnel then utilize the distributed control systems (DCS) for monitoring and controlling the power plant operation. This monitoring approach could be affected by various human factors, including personnel fatigue and improper operation [3]. Furthermore, the alarm mechanism for detecting variables exceeding the fixed thresholds may not detect abnormal states promptly. Hence, it is imperative to implement an effective and efficient anomaly detection system to achieve safe and reliable nuclear power plant operations [4].

Over extended periods of operation, a multitude of sensors within nuclear power plants have generated vast amounts of operational data,

bringing both advantages and challenges for anomaly detection. The historical database includes high-dimensional and long-term sequential data, which may lead to expensive computation and increasing data modality. Meanwhile, interference caused by anomalies and impulse noise makes it difficult to build models of normal state, which brings additional challenges.

The operational condition monitoring of nuclear power plants can be regarded as a task of anomaly detection in multivariate time series, which is a well-established topic in time series analysis. Previous studies on this topic can be classified into three categories: statistical approaches, traditional machine learning methods, and deep learning techniques [5]. Statistical approaches that utilize the 3σ criterion, Z-scores, or Boxplot are useful for detecting anomalies in a univariate stationary time series through analysis of the statistical distribution of data. However, these methods may not be suitable for non-stationary multivariate data. Other widely used statistical-based methods such as autoregressive-based models [6] and Principal Component Analysis (PCA) [7,8] models are commonly used in linear systems, but they may be ineffective when dealing with high volatility nonlinear data. Another

* Corresponding author. College of Science, China Three Gorges University, Yichang, 443002, China.

E-mail address: zsh@ctgu.edu.cn (S. Zheng).

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problem with statistical approaches is they overlook the temporal correlation and dynamics of data, which is essential for time series analysis. Classical machine learning methods tend to derive certain rules for distinguishing normal data and anomalies without assuming a specific generative model. These rules may be based on techniques including clustering (e.g., K-Means [9]), distance calculation (e.g., KNN [10]), density estimation (e.g., LOF [11]), classification (e.g., SVM [12]), and so on. The classical machine learning methods are constrained by sensitivity to parameters, and the optimization problem becomes computationally expensive with the increasing data volume.

The above-mentioned challenges have led to increasing interest and research on solutions based on deep learning networks. Deep learning methods have demonstrated remarkable potential in extracting feature representations for complex structured datasets. Moreover, the predominant state-of-the-art models for time series anomaly detection have confirmed the effectiveness of deep learning techniques. RNN-based models such as LSTM-NDT [13], use Long-Short-Term Memory (LSTM) to predict data of the next time stamp. The prediction residuals will be evaluated with a non-parametric dynamic thresholding approach to detect anomalies. Reconstruction-based approaches such as autoencoder or variational autoencoder models use reconstruction errors or reconstruction probability instead of prediction errors to detect anomalies [14], as autoencoder (AE) approaches, for example. DAGMM [15] is an unsupervised anomaly detection model that utilizes a deep autoencoder to generate a low-dimensional representation and reconstruction errors, while the Gaussian mixture model is used to predict the likelihood. USAD [16] is an autoencoder-based anomaly detection algorithm that utilizes adversarial training. Its autoencoder structure consists of two decoders. These two decoders enhance the model's ability to detect anomalous data through adversarial training, and USAD also demonstrates robustness to Gaussian noise. OmniAnomaly [17] is a Variational Autoencoder (VAE) model aimed at learning the expressive representation of normal data through a stochastic recurrent neural network, and the reconstruction probability is utilized to determine anomalies. Additionally, it proposes an enhanced Peak Over Threshold (POT) method for selecting dynamic thresholds automatically. MAD-GAN [18] uses long short-term memory recurrent neural networks as the generator and discriminator of a Generative Adversarial Network (GAN) to capture the temporal dependency and latent relationships amongst the variables. Prediction errors are used to compute anomaly scores combined with discriminator loss.

As is well known, Transformer-based [19] models are skilled at learning global relevance representation using a multi-head attention mechanism while enhancing efficiency by parallel processing. Tasks of time series forecasting have achieved great progress by utilizing the advantages of Transformer networks for capturing cross-positional dependencies, and it also brings new insights to anomaly detection fields. Anomaly Transformer [20] proposed a new Anomaly-Attention mechanism to compute prior-association and series-association simultaneously. A minimax strategy is conducted to amplify the association discrepancy in model optimization. MT-RVAE [21] proposed a multi-scale Transformer architecture combined with VAE to capture the temporal correlation at different scales and solve the problem with traditional Transformer models that only local information in a certain window is extracted. TranAD [22] employs sequence encoder-decoder architecture with attention mechanisms to capture the temporal trends present in the data and reconstructs the input sequence. The model applies an adversarial training procedure to amplify reconstruction errors during the training phase.

However, the unsupervised Transformer-based anomaly detection models still suffer from interference caused by outliers and anomalies in the training dataset and the originally used mean square error (MSE) loss function is sensitive to these anomalous data, as a result, the reconstruction and anomaly detection performance of the model can be affected. Loss functions based on the maximum correntropy criterion [23] are known to be effective in improving the model's robustness to

outliers and non-Gaussian noise, which can alleviate the interference of anomalous data in training process.

Based on the above motivations, in this paper, we propose a Transformer-based anomaly detection model that utilizes the maximum correntropy criterion to monitor the operational status of nuclear power plants. By introducing a Transformer neural network, the truncated multi-head self-attention mechanism is used to establish the temporal correlation between multivariate data at the present step and previous temporal data over a certain time window. A correntropy loss function based on the maximum correntropy criterion (MCC) is used in training procedures to alleviate the effects of anomalies and outliers in nuclear power datasets. According to the experimental results on public datasets, the proposed *Trans-MCC* model can achieve comparable or superior performance metrics to the baseline models. Moreover, when the training set comprises more abnormal data, the F1 score of *Trans-MCC* model significantly outperformed the *Trans-MSE* without point adjustment. Subsequent experiments utilizing actual nuclear power plant data corroborate the efficacy of the MCC loss function in enhancing model robustness, which is very meaningful for establishing models of normal operational state and improving anomaly detection performance.

2. Related works

As a safety-critical system, nuclear power plants always take operational safety as the top priority, which arouses growing interest in condition monitoring and fault detection of nuclear power plants. Fault detection methods applied in NPPs can be typically categorized into model-based methods and data-driven approaches [1,24].

Model-based methods are a category of approaches that leverage analytical redundancy by employing a system's mechanistic model to characterize its normal operational state. Mechanistic models are typically represented through mathematical equations. Faults could disrupt the normal relationships represented by the mathematical model, leading to statistically significant changes in the residuals between actual sensor measurements and the analytical solutions of the mechanistic model. Hence fault detection can be accomplished by analyzing the residuals statistically. The most widely used model-based anomaly detection approaches include parity equations [25], Kalman filters [26], parameter estimation [27], and diagnostic observers [28]. Godbole et al. [29] utilized the Kalman filters in estimating the non-measurable variables and unknown parameters of a nuclear pool-type research reactor and the approach was available in the estimation of reactor control, protection, instrumentation, and operator information. Khentout et al. [30] employed an analytical redundancy approach to monitor certain parameters of the heat exchanger in the Triga-Mark II nuclear research reactor. They identified instrument faults by comparing the outputs of the system's instruments with model-estimated data. However, model-based anomaly detection methods currently face certain limitations in practical applications within nuclear power plants. The primary reasons include the difficulty of obtaining accurate analytical models for complex systems with numerous sensors, and the model's lack of robustness against noise and disturbances [1].

Compared to model-driven methods, data-driven approaches have received more attention in recent years and have been extensively researched and applied. Data-driven anomaly detection methods primarily rely on data collected under normal operational conditions to establish anomaly detection models using multivariate statistical techniques or machine learning methods. Data-driven approaches have been studied and applied in instrument calibration monitoring, equipment monitoring, reactor core monitoring, and transient identification for NPPs [24]. The most commonly used data-driven methods include Multivariate State Estimation Technique (MSET) [31], Principal Component Analysis (PCA) [32], Artificial Neural Network (ANN) [33], deep learning methods, and so on. Kaistha et al. [34] applied PCA to early fault detection and isolation in nuclear power plant equipment. They achieved this by performing singular value decomposition on the

prediction errors of state and control functions, which allowed them to obtain the fault directions for each specific case. Subsequently, they performed fault isolation by projecting these fault directions. Zavaljevski et al. [35] incorporated a Support Vector Machine (SVM) into the estimation module of MSET, which improved its denoising and generalization characteristics compared to the original MSET method. This enhanced approach was applied to sensor validation and early fault detection in nuclear power plants. Papoikonomou et al. [36] employed a combined LSTM and CNN approach to extract temporal and spatial features from the time-series neutron detector readings, which enabled the detection of the type and source location of reactor perturbation. Ryu et al. [37] proposed an anomaly detection model based on deep autoencoders. This model was trained using normal operational data and relied on the reconstruction errors to discriminate anomalous states. The effectiveness of this method was validated using historical data from the high-flux advanced neutron application reactor (HANARO) and synthetic fault data.

In summary, with the development of the IoT and data mining technologies, data-driven methods have demonstrated significant potential for achieving enhanced anomaly detection performance and gaining more insights into fault-related information. Particularly, the performance advantages of deep learning methods in extracting features from high-dimensional data can effectively address the challenges faced by traditional model-based methods in complex system modeling. However, the impact of noise and disturbances can affect algorithm modeling to a certain extent, necessitating higher demands for algorithm robustness.

3. Methodology

3.1. Problem setup

A nuclear power plant is a typical multi-sensor system, consisting of various kinds of sensors to measure different variables. Due to the different sampling frequencies, the original data collected by different sensors during the same time interval may have varying lengths. To achieve time synchronization and uniform processing, it is common to use down sampling or interpolation to ensure that the data obtained from each sensor is of equal length. Therefore, the nuclear power dataset generated by N sensors within a given time sequence of length T can be represented as a matrix $D \in R^{N \times T}$, in which a column $X = (x_1, x_2, \dots, x_N)^T$ denotes observations of N sensors at a specific timestamp. The process of signal reconstruction can be viewed as trying to seek a reconstruction function f , which can be denoted as:

$$f(x_i) = w^T \varphi(x_i) + b, \quad (1)$$

$\varphi(\cdot)$ is a mapping from the input space to the reconstruction feature space, w denotes for weight vector and b represents the bias term. Therefore, the reconstruction error can be expressed as:

$$\xi_i = w^T \varphi(x_i) + b - y_i. \quad (2)$$

Given a training set $D^{train} = X_t^i|_{i=1}^N, t \leq T$, where X^i is a multivariate sample and T is the length of time series. The anomaly detection task can be converted into building a model through the training dataset D^{train} and utilizing the obtained model to predict the label $C = (c_1, \dots, c_t)$, $c_t \in \{0, 1\}$ of the test set D^{test} , which can indicate whether the tested sample is anomalous.

As the proposed model in this paper, reconstruction-based unsupervised anomaly detection models typically employ data from normal operating conditions as a training set. These models learn the data features that represent normal operation patterns and establish an anomaly detection model based on the assumption that anomalies cannot be effectively reconstructed, therefore the anomalies can be identified based on the values of reconstruction errors.

Fig. 1 illustrates the workflow of the reconstruction-based Transformer anomaly detection model. The model is trained with nuclear power operational data under normal operating conditions and performs anomaly detection based on the reconstruction error of real-time data. It should be noted that unsupervised anomaly detection methods usually assume the training dataset is composed of normal data, but for real-world unlabeled nuclear power data, it is difficult to ensure the normality of the dataset. Therefore, the anomalous data segments as well as outliers may have negative effects on model training. To address the problem, a loss function based on MCC is applied to enhance the robustness of anomaly detection model.

3.2. Transformer model

Following this detection strategy, we incorporate a Transformer network to achieve the reconstruction of input signals. The Transformer is leveraged to extract and learn both the temporal and spatial features of the data under normal operating conditions.

The Transformer architecture utilizes an encoder-decoder structure and employs a multi-head attention mechanism to capture the long-term dependency by parallelly processing the entire sequence, demonstrating enhanced efficiency in handling long sequential data. Unlike recurrent neural networks, as a sequence-to-sequence neural network, it's hard for transformer models to maintain the sequential order directly, which is crucial information for time-series data. To preserve this order, the Transformer network employs positional encoding on the input data before it enters the encoder. The positional encoding function can be expressed as [19]:

$$\begin{aligned} PE_{(pos, 2i)} &= \sin(pos / 10000^{2i/d_{mod\ el}}) \\ PE_{(pos, 2i+1)} &= \cos(pos / 10000^{2i/d_{mod\ el}}), \end{aligned} \quad (4)$$

where $pos \in L$ represents the position of data within the complete sequence of length L , and $i \in d_{mod\ el}$ stands for the dimension of data. The function of the encoder is to encode a set of input vectors and generate a new set of vectors. The vanilla Transformer model employs the self-attention mechanism to accomplish this task. The input vector is multiplied by the randomly initialized weight vector W^Q, W^K, W^V to obtain the query vector Q , the key vector K and the value vector V , and then the corresponding attention score is calculated. The calculation formula is as follows [19]:

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_K}}\right)V, \quad (5)$$

where d_K represents for the dimension of key vector. In the vanilla transformer, the multi-head attention mechanism is conducted to enhance the model's ability to focus on different positions. By using a multi-head attention mechanism, it is possible to establish a correlation between observations at different time stamps.

Fig. 2 is the structure diagram of the transformer-based anomaly detection model. As is shown, the complete sequence of historical operating data is converted to fixed-length time windows as the model input. There are two encoders, one for encoding the input window and another for encoding the output with cross-attention from the former. The decoder output is the reconstruction signal, and the reconstruction error will be sent to the MCC loss function for backward propagation and training procedure.

3.3. Loss function

Correntropy is a generalized correlation function which is first proposed in 2006 based on the cross-information potential in information theoretic learning (ITL) [38]. As a kernel-based local similarity measure [23], correntropy provides a high degree of robustness to outliers and impulsive noise, resulting in its application in signal processing and

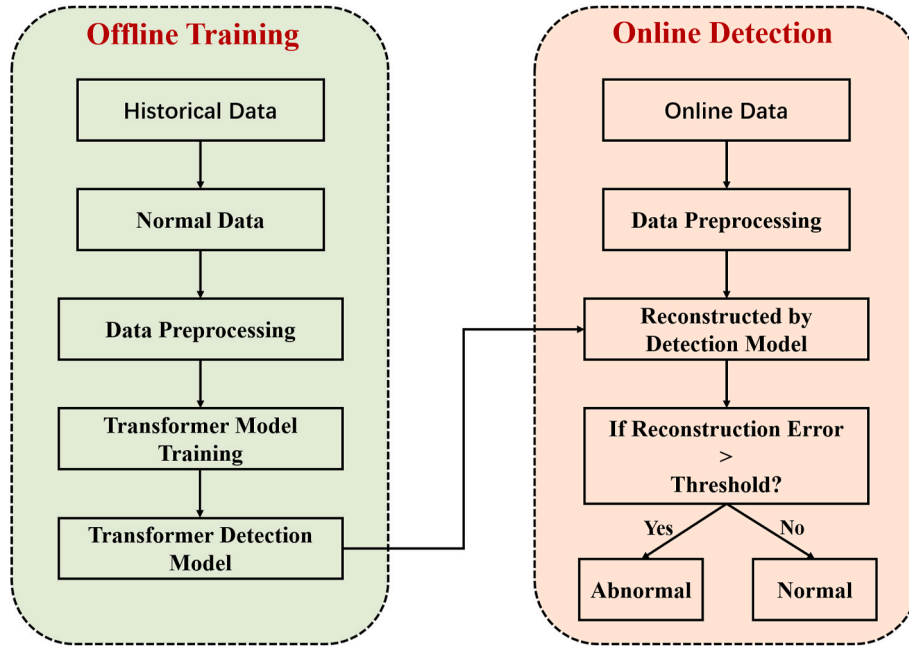


Fig. 1. Workflow diagram of the proposed Transformer-based anomaly detection model.

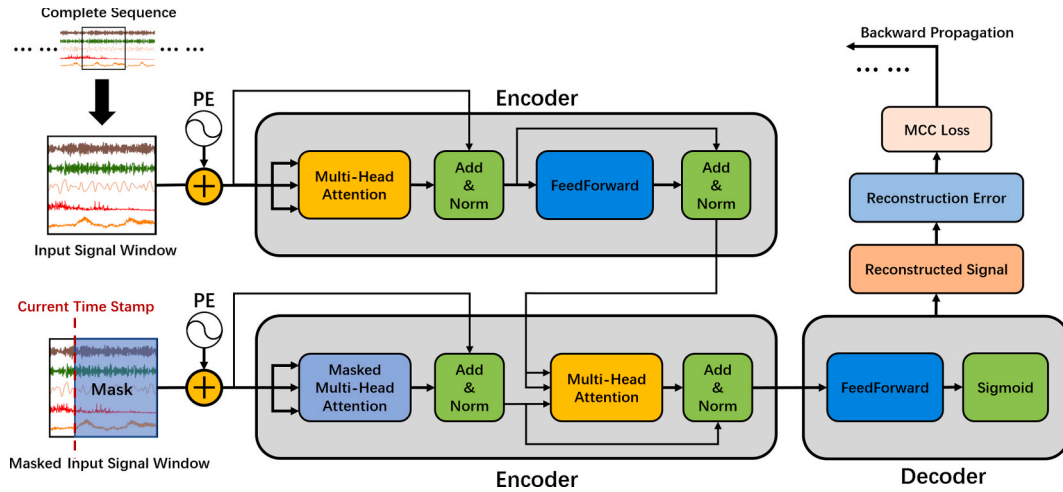


Fig. 2. Structure diagram of Transformer-based anomaly detection model.

machine learning [38–40].

As previously mentioned, the training set of nuclear power plant data lacks label, which leads to the unavoidable influence of outliers and anomalous data on the training process. Numerous deep anomaly detection methods commonly use Mean Squared Error (MSE) as the loss function during training. However, the MSE loss function is prone to large noise and outliers since it is inclined to minimize the maximum error [41], making it challenging to achieve optimization with interference from outliers and anomalies. To alleviate the negative effect of outliers and anomalies, we use a correntropy loss function based on the maximum correntropy criterion.

Generally, correntropy is a local similarity measure between two random variables. For the training process, we use it to measure the similarity between the model's predictions Y with the observations X , as follows [23]:

$$V_{\sigma}(X, Y) = E[K_{\sigma}(X - Y)], \quad (6)$$

where $E[\cdot]$ denotes the exception and K_{σ} is a kernel function. Since the

joint probability density distribution of X and Y are unknown, K_{σ} can be represented as a Gaussian kernel function [23]:

$$K_{\sigma}(x_i, y_i) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{\|x_i - y_i\|^2}{2\sigma^2}\right). \quad (7)$$

Thus, given a set of $D = \{(x_i, y_i)\}_{i=1}^N$, the correntropy between observations X and predictions Y can be expressed as:

$$V_{\sigma}(x_i, y_i) = \frac{1}{N} \sum_{i=1}^N \left[\frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{\|x_i - y_i\|^2}{2\sigma^2}\right) \right]. \quad (8)$$

For the deep learning training procedure, the correntropy induced loss function L is defined as:

$$J_{MCC}(X, Y) = C(1 - V_{\sigma}(x_i, y_i)), \quad (9)$$

where $C = [1 - \exp(-\frac{1}{2\sigma^2})]^{-1}$ is a scaling constant.

The transformer model utilizes a backpropagation-based stochastic gradient descent algorithm to optimize. For the MSE loss function, the

optimization problem can be expressed as:

$$\min J_{MSE} = \frac{C}{2} \sum_{i=1}^N \xi_i^2 + \frac{1}{2} \|W\|^2. \quad (10)$$

For the correntropy loss function, the maximum correntropy criterion achieves optimization by maximizing the correntropy as follows:

$$\max J_{MCC} = \frac{C}{2} \sum_{i=1}^N \exp\left(-\frac{\xi_i^2}{2\sigma^2}\right) - \frac{1}{2} \|W\|^2, \quad (11)$$

to meet the requirements of gradient descent methods, Eq. (8) can be transformed into the following form:

$$\min J_{MCC} = 1 - \frac{C}{2} \sum_{i=1}^N \exp\left(-\frac{\xi_i^2}{2\sigma^2}\right) + \frac{1}{2} \|W\|^2. \quad (12)$$

Fig. 3 depicts the variation curves of the MCC and MSE loss functions with respect to the errors. As shown in Fig. 3, the loss value of MSE loss function increases rapidly in a quadratic manner with the increase of error. Thus, the model trained with the MSE loss function may be prone to large impulse noise and outliers since it depends on minimizing the mean square error on training samples. While the MCC loss function is a local similarity measurement which can amplify the change of losses within a certain range of errors, without causing excessively large changes in loss for large errors outside of the range. Hence the MCC loss function can alleviate the negative effect of outliers and anomalies with large errors.

In addition, the mentioned error range depends on the value of bandwidth σ in the Gaussian kernel, the value of σ in this study is selected according to the empirical rule proposed by Silverman [42] as below:

$$\sigma = 0.9AN^{-\frac{1}{5}}, \quad (13)$$

in which $A = \min\{std(data), quantile(data)/1.34\}$, and N stands for the data length.

During Transformer model training, the input sequence is encoded first. Thereafter, the encoded representation and partial target sequence details are utilized for decoding and obtaining the predicted sequence. The difference between the predicted and true sequences is calculated for computing the gradient of the relevant entropy loss function with respect to each parameter. Finally, the gradient descent optimization algorithm is leveraged to update the parameters gradually to reduce the loss function.

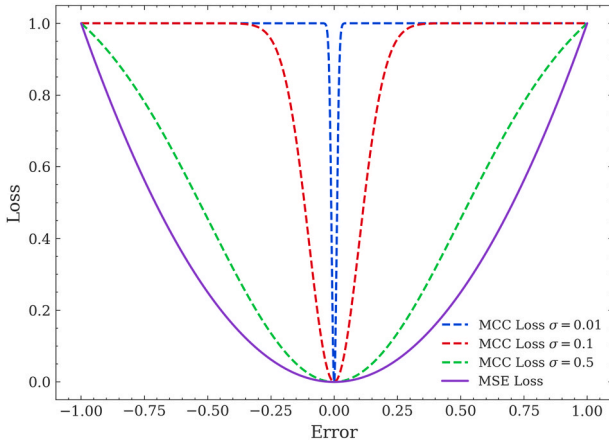


Fig. 3. The MCC and MSE Loss curves via errors (horizontal-axis: normalized error; vertical-axis: normalized loss), the σ of MCC loss function is in $\{0.01, 0.1, 0.5\}$.

4. Experiments

To validate the effectiveness of the proposed model for anomaly detection in nuclear power plant operational data, we conducted three comparative experiments. First, given the scarcity of fault condition samples in real nuclear power plant operational data [43], we generated simulated datasets for different fault conditions using a simulator. The effectiveness of the transformer-based model is proved by comparing it with the other three baseline anomaly detection models on the labeled simulated datasets. Second, the real nuclear power plant operational dataset may contain noise and disturbances, which may have negative impacts on model building of normal condition data. Therefore, to assess the robustness of the *Trans*-MCC model in the presence of anomalous data in the training set, we introduced different proportions and levels of impulse noise into the simulated normal condition training data. This allowed us to compare the detection performance of *Trans*-MCC with the original *Trans*-MSE, thus verifying whether *Trans*-MCC effectively enhances the model's robustness in the presence of abnormal data in the training set. Finally, since the characteristics of real nuclear power plant operational data are different from simulated data generated by the simulator to some extent, we selected a real-world anomalous data instance to validate the model's detection capabilities and its robustness against anomalous disturbances in real nuclear power operation data.

In this section, the simulation datasets used in the experiments and the adopted evaluation metrics are introduced first. Subsequently, we analyzed the results obtained from the comparative experiments on the simulated dataset, which encompassed tests of anomaly detection performance and robustness to anomalous training sets. Finally, we assessed the application of the proposed model on real-world nuclear power plant data.

4.1. Datasets and metrics

The simulated dataset was generated using a 1000 MW nuclear power plant full-scope simulator, which can generate real-time operational data and simulate various fault conditions [44]. Normal condition samples and various kinds of fault condition samples were generated by the simulator.

Due to the vast number of various sensors in nuclear power plants, in the experiment section of the simulated dataset, we focused on the research of Steam Generator (SG). As an essential component situated between the primary and secondary loops, SG plays a critical role in nuclear power plants by facilitating heat exchange and steam generation. The liquid level within the steam generator is a vital parameter for assessing the operational state of the nuclear power plant. We selected certain sensor parameters related to the liquid level of SG-1 under 100% FP steady-state conditions. Furthermore, we generated four fault conditions using the simulator, which could potentially impact the steam generator's liquid level.

The accident dataset consists of simulated operational data of four different accident conditions and was employed for validating the effectiveness of the proposed model. These conditions include Primary Feedwater Pipe Rupture (PFPR), Steam Line Break Inside Containment (SLBIC), Steam Generator Tube Rupture (SGTR), Steam Generator Secondary Side Leakage in the Hot-Leg (SG2L-HL), and Steam Generator Secondary Side Leakage in the Cold-Leg (SG2L-CL). The accident severity is set as 0.01, primarily based on two considerations. First, a certain level of fault severity may trigger shutdown or reactor trip, therefore setting the severity value too high would result in a limited number of available samples. Additionally, smaller deviations in operation states require more stringent detection by the model, making it more reflective of differences in the detection performance of every model. As a result, the data samples used for training and validating the anomaly detection model of the nuclear power plant encompass one normal condition and four accident conditions. The statistics of the simulated fault conditions are shown in Table 1, including the mean (μ)

Table 1

Statistics of selected operation parameters in the simulated accident datasets.

Parameters	Normal Condition		PFPR		SLBIC		SGTR		SG2L	
	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ
SG1 Liquid Level WR	−1.4714	0.0007	−1.4813	0.0206	−1.4647	0.0068	−1.4706	0.0018	−1.4436	0.0183
SG1 Liquid Level NR	−0.0025	0.0008	−0.0153	0.0291	−0.0007	0.0045	0.0003	0.0015	−0.0016	0.0087
Feed Water Flow Rate	1905.66	1.0131	1932.25	22.65	1918.86	9.2572	1905.19	1.0608	1892.34	18.737
Feed Water Temperature	222.20	0.0203	222.18	0.0333	222.15	0.0809	222.22	0.0025	221.89	0.3892
Steam Flow Rate	1896.78	0.9548	1899.50	1.5787	1910.01	9.2573	1897.81	0.2440	1883.48	17.908
Steam Pressure	6.7219	0.0012	6.7231	0.0006	6.6940	0.0317	6.7274	0.0041	6.6413	0.0798
S–W Pressure Difference	0.8898	0.0006	0.8899	0.0047	0.8897	0.0045	0.8902	0.0017	0.8910	0.0137
Turbine Load	98.7202	0.0456	98.8343	0.0654	98.85	0.1144	98.7511	0.0048	98.1744	0.7183

and standard deviation (σ) of selected variables under different fault conditions.

The dataset comprises operation data of SG-relevant parameters generated by the simulator, and we partitioned them into training and test sets based on the operational states, which follow the detection strategy of unsupervised reconstruction-based models. Table 2 provides statistics of the simulation dataset used in experiments. The normal condition data was used as training data while the four accident datasets were taken as test datasets.

To evaluate and compare the performance of different detection models quantitatively, we employ Precision, Recall, and F1 scores. The calculation formulas for each metric are shown as follows:

$$\begin{aligned}
 P &= \frac{TP}{TP + FP} \\
 R &= \frac{TP}{TP + FN} \\
 F1 &= \frac{2 \times P \times R}{P + R}
 \end{aligned} \quad (14)$$

The above formula indicates that precision, defined as the proportion of true abnormal points among all points detected as abnormal, represents the true positive rate and higher precision values correspond to a lower rate of false alarms. On the other hand, recall, defined as the proportion of true abnormal points identified among all actual abnormal points, reflects the sensitivity of the model with higher recall values indicating lower rates of missed alarms. The F1 score is a widely used metric in evaluating the performance of models, as it provides a balanced measure of both precision and recall. Another thing to note is that an evaluation technique known as point adjustment (PA) [45] is more often used in the benchmark of anomaly detection models. Under this technique, if a single point within a segment has been identified as abnormal, the whole segment will be considered as abnormal. The use of PA has been shown to have a significant impact on the evaluation metrics of anomaly detection models [14]. For a more objective model comparison, we adopt evaluation metrics without PA.

Since the variables of nuclear power operation data are reconstructed using Channel-Dependent approaches, all the evaluation metrics are calculated by each channel, and the final evaluation metrics are the average of the metrics of all channels.

Due to the evaluation metrics being derived by comparing anomaly scores to the thresholds to generate category labels, the selection of the thresholds greatly impacts the evaluation metrics. To facilitate a fair comparison, we adopted the Peak Over Threshold (POT) [46] method to

choose suitable thresholds for each model. The threshold selection method involves determining the threshold by examining the statistical distribution of the reconstruction losses from the normal condition data, and reconstruction losses higher than the thresholds would be identified as anomalies.

4.2. Results and analysis

To verify the detection performance of a transformer-based model and the effectiveness of the MCC loss function in enhancing robustness to anomalies in the training set, a series of comparative experiments was carried out. We compared the proposed model's detection performance to that of some competitive baseline models, such as DAGMM [15], USAD [16], and OmniAnomaly [17]. The model performance comparison experiment was conducted on the above-mentioned simulation nuclear power operation datasets.

As shown in Fig. 4, the simulated PFPR accident has an initial timestamp of around 2014. Data before this timestamp represent normal operating conditions and exhibit characteristics consistent with the training dataset, resulting in minimal reconstruction error. After the insertion of the fault, the equipment state deviates from normal operation on account of changed parameters, leading to significantly higher reconstruction errors compared to normal conditions.

It's worth noting that this fault was canceled at a timestamp of about 2915. After the fault cancellation, parameters gradually returned to normal conditions. However, data collected before complete recovery still differed from normal conditions and were considered as anomalies. The actual anomaly labels were manually annotated based on data statistical characteristics.

A comprehensive analysis of the detection results for various variables allows us to draw the following conclusions. Firstly, the detected starting timestamp of the fault closely aligns with the ground truth of the starting timestamp, demonstrating the effectiveness of the proposed model in promptly detecting anomalies. Subsequently, samples during the fault's duration were almost totally identified as anomalies. The Intersection over Union (IoU) between detected anomaly segments and actual anomaly segments is high, indicating the model's ability to recognize entire anomaly events, unlike some of the anomaly detection models that can only identify a few anomalous points. Nevertheless, false detections do occur to a certain degree, the detected duration of anomalies for some variables was longer than the labeled true anomaly segments. This can be attributed to the data labeling and the selection of alarm thresholds. The evaluation metrics of all 8 variables were

Table 2

Statistics of the comparative experiments on simulation datasets.

Dataset	Basic Operational Status	Accident types	Accident Severity	Total Size	Anomalies	Anomaly Rate
Training Set	100% FP Steady-State	Normal	/	30000	0	0
Test Set	100% FP Steady-State	PFPR	0.01	4418	1976	44.73%
		SLBIC	0.01	4560	1960	42.98%
		SGTR	0.01	4438	1832	41.28%
		SG2L	0.01	5076	2024	39.87%

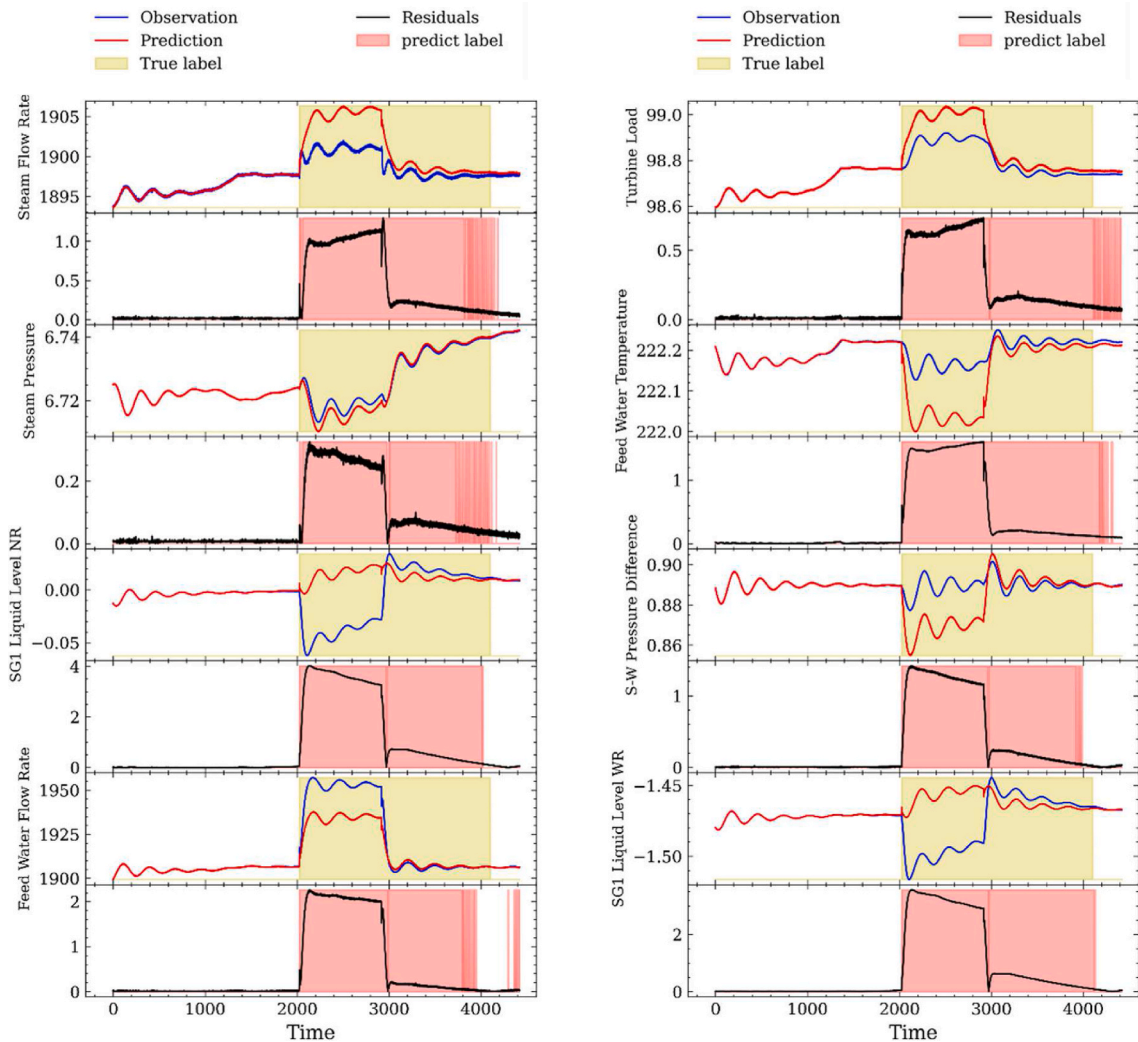


Fig. 4. The anomaly detection results of *Trans*-MCC model on the PFPR accident dataset.

Table 3
Channel-independent evaluation metrics of the *Trans*-MCC model on four accident datasets.

Variables	PFPR			SLBIC			SGTR			SG2L		
	F1	P	R	F1	P	R	F1	P	R	F1	P	R
SFR	0.9645	0.9868	0.9431	0.8848	0.9987	0.7943	0.9612	0.9321	0.9922	0.8996	0.9747	0.8353
SP	0.9318	0.9988	0.8733	0.9304	0.8706	0.9990	0.7967	0.8999	0.7148	0.8485	0.8541	0.8430
SG1 LL NR	0.9775	1	0.9561	0.9272	0.8664	0.9971	0.9741	0.9776	0.9706	0.9349	0.8781	0.9995
FWFR	0.9331	0.9839	0.8872	0.7645	0.8228	0.7138	0.7973	0.8327	0.7649	0.8120	0.8369	0.7885
TL	0.9622	0.9275	0.9995	0.7481	0.8173	0.6897	0.9286	0.8668	1	0.9255	0.8661	0.9937
FWT	0.9712	0.9440	1	0.9276	0.8665	0.9980	0.9272	0.8664	0.9971	0.9277	0.8666	0.9981
SWPD	0.9630	1	0.9287	0.9468	0.8997	0.9990	0.9162	0.8461	0.9990	0.9230	0.8577	0.9990
SG1 LL WR	0.9935	0.9894	0.9975	0.9239	0.8601	0.9980	0.9775	0.9674	0.9879	0.9391	0.8859	0.9990
Average	0.9621	0.9788	0.9482	0.8817	0.8753	0.8986	0.9098	0.8986	0.9283	0.9013	0.8775	0.9320

calculated, and the average F1, precision, and recall are taken as the final evaluation metrics of the detection model. Table 3 shows the channel-dependent evaluation metrics of the *Trans*-MCC model on all four accident datasets.

Simultaneously, we conducted experiments for other three state-of-the-art (SOTA) models: DAGMM, USAD, and Omnianomaly, all of which were applied to four distinct fault datasets. As presented in Table 4, the *Trans*-MCC model achieved superior or comparable

Table 4
Comparison of evaluation metrics on four accident datasets.

Model	PFPR			SLBIC			SGTR			SG2L		
	F1	P	R	F1	P	R	F1	P	R	F1	P	R
USAD	0.9163	0.8975	0.9415	0.9233	0.8998	0.9591	0.9130	0.8905	0.9409	0.8923	0.8266	0.9714
DAGMM	0.8701	0.8701	0.8930	0.8817	0.8753	0.8986	0.8935	0.8748	0.9193	0.8143	0.9027	0.7778
OmniAnomaly	0.8937	0.8772	0.9152	0.9008	0.9107	0.9142	0.9278	0.9216	0.9438	0.8499	0.8137	0.9073
<i>Trans</i> -MCC	0.9621	0.9788	0.9482	0.9235	0.8778	0.9757	0.9366	0.8850	0.9956	0.8693	0.8151	0.9362

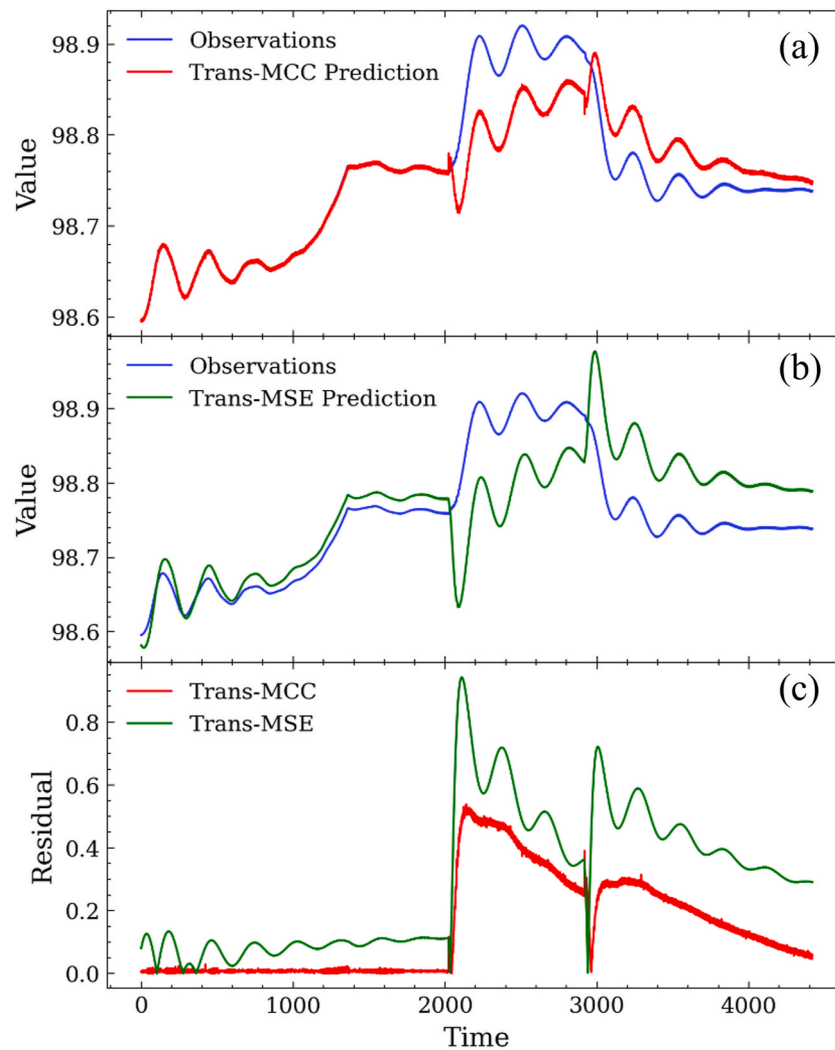


Fig. 5. Comparison of reconstruction performance between the TranAD-MSE model and TranAD-MCC model.

detection performance across all four fault datasets. This validates the efficacy of the proposed Transformer-based model for detecting faults generated by the simulator. These results align with the performance of other Transformer-based anomaly detection models, such as Anomaly Transformer [20] and TranAD [22]. It suggests that the Transformer model holds the potential for detecting anomalies in time series and merits further research.

As has been discussed above, reconstruction-based unsupervised anomaly detection models tend to use normal operation data for training. However, due to the absence of data labels and the presence of noise and disturbances, in real-world scenarios, datasets used for model training inevitably contain outliers and anomalous segments. These anomalous samples can adversely affect the model establishment, particularly for the original Transformer model utilizes the MSE loss function, which is sensitive to outliers and anomalies. This sensitivity hampers the model's ability to effectively learn the features of normal operational data, consequently impacting detection performance. To validate the effectiveness of the MCC loss function, we designed comparative experiments in which we introduced impulse noise, or outliers, at varying proportions into the dataset representing normal operational conditions. We then compared the detection performance of the *Trans-MCC* model, which employs the MCC loss function, with that of the original *Trans-MSE* model.

In Fig. 5, subgraph (a) and (b) present the reconstructed signal of the *Trans-MCC* and *Trans-MSE* model on the PFPR accident dataset,

respectively. The reconstruction error of the two models is presented in subgraph (c). The training set consisted of normal operation data was injected with impulse noise. Both the *Trans-MCC* and *Trans-MSE* models were trained using the training set that now included anomalous data. Subsequently, they were tested on the fault data without any additional outliers. As shown in Fig. 5, even though both models exhibited larger reconstruction errors during fault periods compared to normal operational periods, it can still be observed that *Trans-MSE* is more susceptible to the influence of anomalous data. This is evidenced by *Trans-MSE* having higher reconstruction errors for normal data compared to *Trans-MCC*. With the alarm threshold unchanged, this could lead to an increase in false positives and a decline in detection performance.

To further investigate the impact of noise and disturbances on the training of normal operating condition models, we introduced pulse noise into the training dataset at varying proportions relative to the overall training data. We then conducted testing on the PFPR fault datasets, and the test results are presented in Table 5.

From Tables 5 and it can be observed that as the proportion of pulse noise increases, the performance metrics of all models show varying degrees of degradation. This can be attributed to two main factors. First, noise influences model training, leading to increased reconstruction errors in normal data in the test set, which can result in some false positives. Furthermore, since the anomaly detection threshold is determined based on the POT method of normal operational data, excessive noise can also affect the threshold, leading to a decrease in detection

Table 5

Comparison of *Trans*-MCC and *Trans*-MSE on the PFPR dataset with different levels of injected impulse noise in the training set.

Model	Anomaly Rate	PFPR		
		F1	P	R
<i>Trans</i> -MSE	1%	0.9138	0.9213	0.9065
	5%	0.8293	0.8812	0.7832
	10%	0.7554	0.8563	0.6758
<i>Trans</i> -MCC	1%	0.9284	0.9461	0.9114
	5%	0.8893	0.9231	0.8578
	10%	0.8110	0.8760	0.7549

metrics. Through the comparison of the results between *Trans*-MCC and *Trans*-MSE, we can observe that *Trans*-MCC consistently outperforms *Trans*-MSE, especially when the proportion of anomalies increases beyond a certain threshold. This difference becomes more pronounced as the anomaly proportion rises. The primary reason for this improvement is that the MCC loss function mitigates the interference caused by outliers during model training by suppressing excessively large losses. In contrast, the MSE loss function tends to reduce the maximum loss during training, making it more susceptible to interference.

4.3. Application on real nuclear power data

4.3.1. Detection performance on real operation dataset

As is well known, the data characteristics of real nuclear power data may be different from simulated data generated by a simulator to some extent. This discrepancy primarily arises from the presence of noise and disturbances in the real environment, along with variations in the response times of different parameters. To validate the performance of *Trans*-MCC on real data, we selected historical data from sensors related to the charging pump RCV001PO from the chemical and volumetric control system in a nuclear power plant. The dataset consists of measurements of 30 variables, including temperature, current, and vibration signals. The data contained an instance of a pump trip anomaly and the anomalous segments were labeled manually to facilitate quantitative evaluation of the model's performance. Statistical details about the real dataset are presented in Table 6.

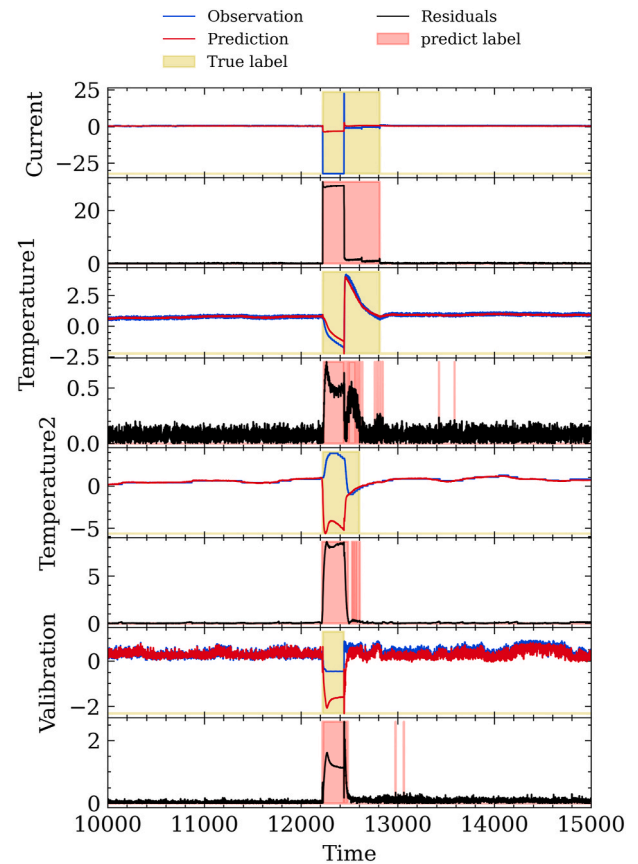
Fig. 6 displays partial detection results of the *Trans*-MCC model on the real dataset, which include one current signal, two temperature signals, and one vibration signal. The results indicate that the model can promptly detect the occurrence of faults, with the timestamps for the detected faults being nearly consistent with the actual fault occurrence times. Furthermore, the reconstruction errors during the fault occurrence period are significantly larger than those during normal operating periods, confirming the effectiveness of the proposed method in detecting abnormal events. However, it's important to note that the model does exhibit some false positives and false negatives. In practical applications, the balance between these two aspects needs to be tailored to specific requirements.

In the comparative experiments on the real nuclear power dataset, the proposed Transformer-based method was compared not only with the three previously mentioned SOTA deep learning methods but also with the classical PCA method, which is widely used in nuclear power and other industrial applications. Table 7 presents the experimental

Table 6

Statistics of real nuclear power operation dataset of the charging pump RCV002PO.

Dataset	Basic Operational Status	Dimensions	Total Size	Anomalies	Anomaly Rate
Train Set	85% FP Steady-State	30	39120	/	/
Test Set			29000	595	2.05%

**Fig. 6.** Comparison between the TranAD-MSE model and TranAD-MCC model.

results for each method, listing the average values of the channel-specific metrics.

Based on the results in Table 7, we make the following observations. Firstly, all the methods successfully detect the occurrence of the pump trip fault, the detection performance of the five deep learning methods is relatively close, while the performance of the PCA method is comparatively poor. This discrepancy can be primarily attributed to PCA's weak performance in detecting vibration signals, which drags down the overall performance. Sequencely, it's noticeable that different models identify the timestamps of the fault onset differently. This is primarily due to variations in reconstruction performance, leading to different threshold values. Models with better reconstruction performance exhibit smaller reconstruction errors for normal operating data, which in turn results in smaller threshold values. This allows them to detect faults when even minor changes occur. Conversely, models with poorer reconstruction performance yield larger reconstruction errors for normal operating data, resulting in larger threshold values. Therefore, fault data needs to exhibit significantly higher reconstruction errors to be classified as anomalies, causing a delay in detecting the onset of the fault. Such delays can be detrimental to the timely handling of faults.

Table 7

Experiment results of performance comparison on real nuclear power operation data.

Model	RCV002PO		
	F1	P	R
DAGMM	0.8082	0.8958	0.7541
USAD	0.8780	0.9389	0.8482
OmniAnomaly	0.8743	0.9350	0.8418
PCA	0.5080	0.8386	0.3914
<i>Trans</i> -MSE	0.8680	0.9278	0.8154
<i>Trans</i> -MCC	0.8797	0.9310	0.8632

Furthermore, the performance of *Trans*-MSE is quite similar to *Trans*-MCC, primarily because the training set mainly comprises normal data without interference from impulse noise or anomalies. To validate the advantage of *Trans*-MCC in terms of robustness, additional experiments using real data from the RCV002PO charging pump is conducted in the next section. The dataset contains outliers and disturbances that can potentially impact model training.

4.3.2. Robustness analysis of the detection model

As previously mentioned, real nuclear power plant data contains noise and external disturbances that can affect the modeling of normal operating modes. Here we take the data of the RCV200MT temperature sensor from the RCV002PO charging pump as an example. In the real temperature signal curve in Fig. 7, we can observe that the temperature signal contains outliers and anomalous segments due to interference and impulse noise. These anomalies may not necessarily be attributed to equipment faults, but they do exhibit distinct data characteristics compared to the normal condition data. Therefore, measures need to be

taken to make the detection model robust to impulse noise and abnormal disturbances. The introduction of the MCC loss function is intended to solve this problem.

As shown in Fig. 7, the *Trans*-MSE model is more likely to be affected by impulsive noise and anomalies, while MCC loss function with suitable σ value can alleviate the negative impacts. This is of great significance for monitoring the actual operational conditions of nuclear power plants. When an abnormal state occurs, the reconstruction error of the model is greater compared to the MSE model. Thus, the abnormal state can be detected at an earlier stage before triggering the preset thresholds of the power plants, which is helpful in achieving early warning and processing.

To further investigate the impact of the MCC loss function in model training, a comparison experiment was conducted on the test set between *Trans*-MSE and *Trans*-MCC with different σ values. Fig. 8 presents the prediction signals of *Trans*-MCC models with different σ values. It can be found that if $\sigma = 0.5$, the reconstruction signal is very similar to *Trans*-MSE model's result, and if $\sigma = 0.01$, the model can hardly learn

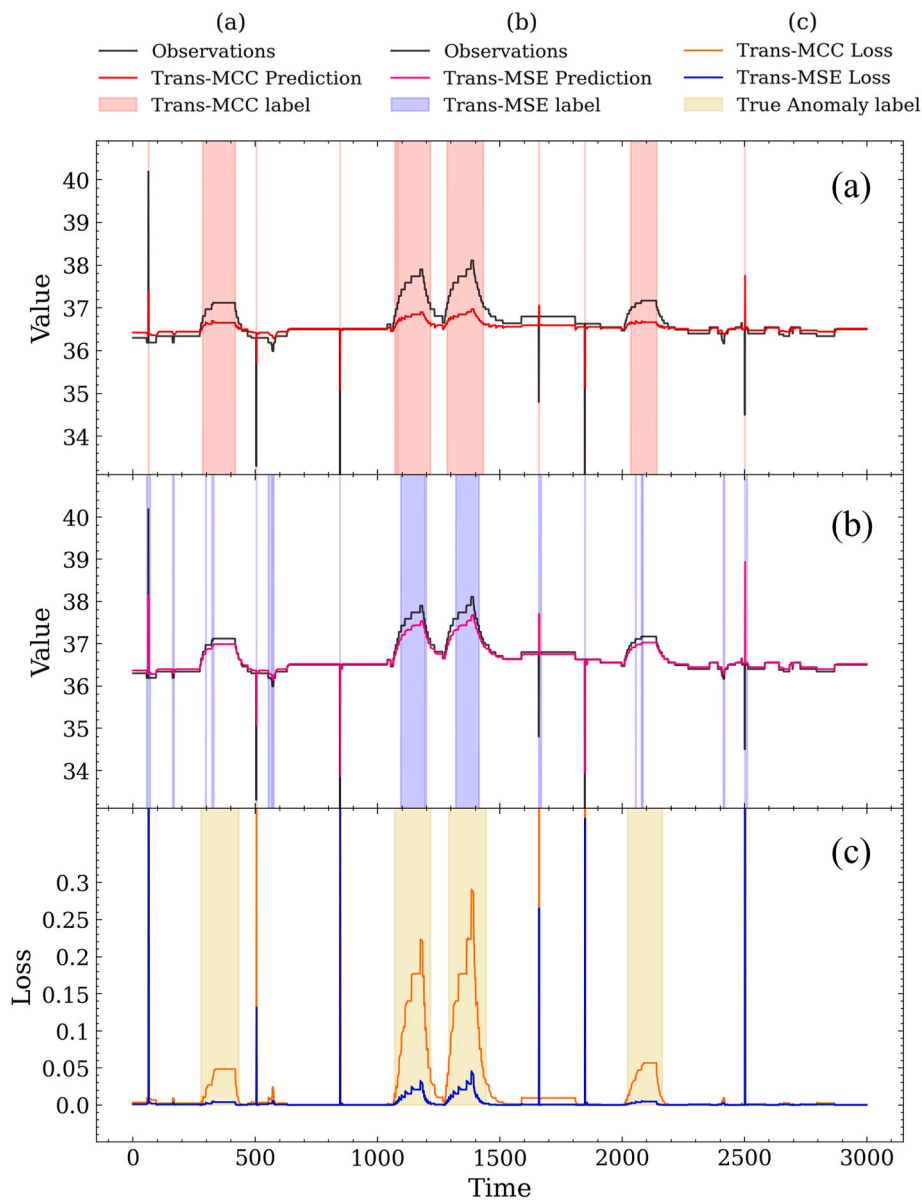


Fig. 7. Experimental results on RCV002PO-200 MT data. (a) and (b) are the comparison between observations and predictions by *Trans*-MCC model and *Trans*-MSE model, respectively. The predicted anomalies are highlighted in red and blue. (c) Is the reconstruction loss of the two models, the true anomaly label is highlighted in yellow. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

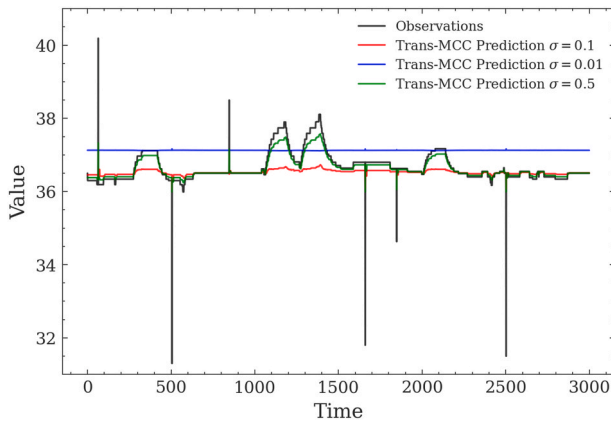


Fig. 8. Experimental results on RCV002PO data. (Horizontal-axis: time stamps; vertical-axis: the observation value and predict value of temperature), the σ of MCC loss function is in $\{0.01, 0.1, 0.5\}$.

Table 8

Experiment results of performance comparison on real nuclear power operation data.

Model	RCV002PO						
	TP	TN	FP	FN	P	R	F1
Trans-MSE	262	2351	52	335	0.8344	0.4389	0.5752
Trans-MCC ($\sigma = 0.01$)	6	2397	6	591	0.4999	0.0101	0.0197
Trans-MCC ($\sigma = 0.1$)	545	2347	56	52	0.9068	0.9129	0.9098
Trans-MCC ($\sigma = 1$)	278	2351	52	319	0.8424	0.4657	0.5998

any features of the temperature signal. While if the value of σ is suitable, just as $\sigma = 0.1$, the model can learn features of normal data and be robust to anomalous data at the same time.

Table 8 shows the evaluation metrics of *Trans-MSE* and *Trans-MCC* model with different σ , it should be noted that the performance metrics are highly correlated with the selection of alarm thresholds. After repeated experimental verification, we chose to select the threshold according to the 3-sigma principle of the reconstruction loss of each model on normal data (anomalous data segments have been removed), all the reconstruction losses are calculated by means of MSE. This method of threshold selection has been experimentally proved to be relatively fair for different models. According to the results in Table 8, *Tran-MSE* model was affected by anomalous data evidently, a lot of anomalous data segments are falsely alarmed. In *Trans-MCC* models, when $\sigma = 0.5$, the detection performance is similar to *Tran-MSE*, which is consistent with the previous analysis; when $\sigma = 0.01$, the prediction signal is almost a straight line, and the result is not available; While when $\sigma = 0.1$, the performance is significantly improved, the model shows robustness to anomalous data with large errors, *Trans-MCC* model is able to outperform *Trans-MSE* by increasing F1 score by up to 0.31. The chosen of σ value is depending on the distribution of normal data, which has been discussed in the front section. For operational state monitoring of nuclear power plants, the technicians can choose the sigma value that best meets the needs of daily monitoring to achieve the best balance of false alarms and missing alarms.

Another advantage of MCC-based model is the faster training procedure, Fig. 9 present the loss decline curves of *Trans-MCC* and *Trans-MSE* model, it can be found that *Trans-MCC* model convergence faster than *Trans-MSE* model due to the error weighting by MCC loss function. This may be meaningful for the self-updating and self-learning for real-time detection since faster training means shorter response time.

The effects of the MCC loss function can be concluded as enhancing robustness to anomalies and outliers in the training set, which may lead

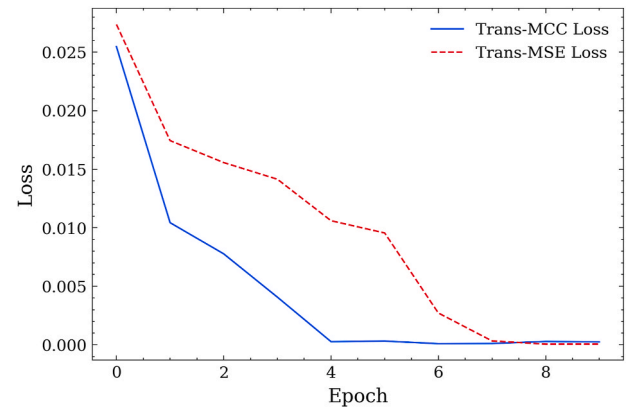


Fig. 9. Loss decline curves of *Trans-MCC* and *Trans-MSE*.

to better detection performance of the detection model, such as lower missing alarm rate and earlier detection of anomalous conditions.

5. Conclusion

The anomaly detection of nuclear power operating data is an important part of the digital development of nuclear power plants with the increasing demands of automatic condition monitoring. This work proposed a transformer-based anomaly detection model trained with the MCC loss function to enhance the robustness of the proposed model. The *Trans-MCC* model can achieve the same and even better performance than baseline methods according to the results of comparison experiments on the simulation datasets of different types of faults. Further experimental results on the real nuclear power operation dataset illustrate the effectiveness of the detection model on real data, and the MCC loss function is proven can alliterate the interference by disturbances and impulsive noise in the training set. The F1 score of *Trans-MCC* on a specific dataset is improved by up to 0.31 than *Trans-MSE* model. By selecting a suitable bandwidth, the *Trans-MCC* model can achieve excellent detection performance without being interfered by anomalous data in model training, which is very meaningful for real-world anomaly detection of nuclear power plants.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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